MedVQA-Router: Bridging Binary Classification and Open-Ended Captioning for Medical Visual Question Answering

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Abstract

Medical Visual Question Answering (Med-VQA) requires simultaneous understanding of clinical images and natural-language questions. We address both the binary (yes/no) and open-ended subsets of Med-VQA using the `robailleo/medical-vision-llm-dataset` [9], which contains 4,793 samples across X-ray, CT, MRI, and ultrasound modalities. Our two-stage pipeline routes questions via a rule-based classifier, directing binary questions to a multimodal fusion model (ResNet-18 + BioBERT) and open-ended questions to GPT-5 Mini via the OpenAI API. For binary VQA we compare text-only baselines (TF-IDF + LR/SVM), an image-only CNN (ResNet-18), and fusion models. Our best binary classifier—ResNet-18 fused with fine-tuned BioBERT—achieves **79.5% accuracy** and **0.83 AUC-ROC**, outperforming single-modality baselines by over 7 percentage points.

1 Introduction

Medical Visual Question Answering (VQA) has attracted significant research interest as a tool for automated clinical decision support. A Med-VQA system must jointly reason over a clinical image and a natural-language question to produce a correct answer—either binary ("Is there a fracture?") or open-ended ("What abnormality is visible?"), requiring different modelling strategies.

Medical datasets are limited in size due to the high cost of expert annotation and privacy regulations. Clinical images exhibit significant visual diversity across modalities, while questions use a highly specialized vocabulary. Standard vision-language models pretrained on natural images often fail to capture subtle pathological markers critical for accurate diagnosis.

We propose a two-stage pipeline that prioritizes task specialization: a routing classifier separates binary from open-ended questions, followed by a multimodal fusion architecture combining fine-tuned ResNet-18 with domain-specific BioBERT embeddings. This approach achieves 79.5% accuracy and 0.83 AUC-ROC on binary classification, significantly outperforming general-purpose models like CLIP.

2 Background

Visual and Medical VQA. Since the VQA v2 dataset [2], VQA research has evolved from monolithic architectures to complex attention-based mechanisms. In the medical domain, datasets such as VQA-Med [4] and SLAKE [7] benchmark performance on radiology and pathology images, where accuracy is safety-critical and linguistic bias is a known challenge.

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Transfer Learning and Biomedical Language Models. Due to the scarcity of annotated medical images, transfer learning from ImageNet-pretrained ResNet architectures [3] is standard practice. BERT [1] has spawned specialized variants such as BioBERT [5], pretrained on PubMed abstracts and PubMed Central full texts, making it significantly better suited to clinical terminology than general-domain models.

Vision-Language Models. CLIP [8] aligns images and text in a shared latent space via dual encoders, while BLIP [6] enables flexible generation through a multimodal encoder-decoder mixture. GPT-5 Mini extends this to large-scale instruction-following captioning via the OpenAI API.

3 Data

3.1 Dataset

We use the `robailleo/medical-vision-llm-dataset` [9] from HuggingFace, a combined medical vision-language dataset comprising **4,793 samples** drawn from two sources: ROCO (3,000 samples) and VQA-RAD (1,793 samples). The official split yields **3,834 training** and **959 validation** samples.

Modality distribution: X-ray dominates with 2,325 samples (48.5%), followed by CT (1,351; 28.2%), unknown modality (812; 16.9%), MRI (231; 4.8%), ultrasound (70; 1.5%), and microscopy/endoscopy (4; $<0.1\%$). **Body-part distribution:** Unknown (2,856), chest (718), abdomen (468), head (408), extremities (125), spine (119), and pelvis (99).

3.2 Preprocessing and Split

Images are resized to 224×224 and normalized using ImageNet statistics. Grayscale images are converted to three-channel RGB; questions are tokenized to a maximum of 128 tokens. After routing, the binary subset contains **940 samples** (750 train / 190 val), nearly balanced at 49.5% / 50.5% yes/no. The open-ended subset contains the remaining samples.

4 Methods

Our system has three components: (1) a routing classifier, (2) a binary VQA pipeline, and (3) an open-ended captioning pipeline.

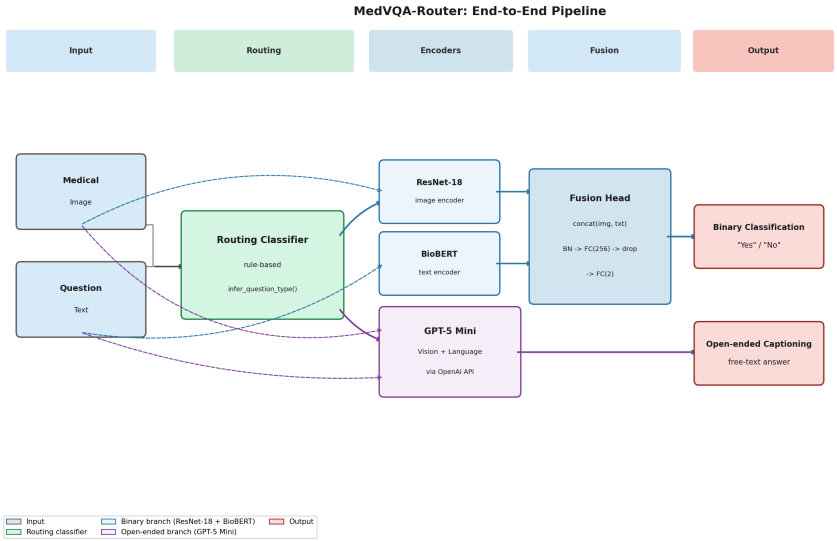

Figure 1: End-to-end pipeline: the routing classifier directs inputs to the binary classification or open-ended captioning branch.

4.1 Routing Classifier

A rule-based router in `src/utils.py` inspects each answer: "yes" or "no" answers are directed to binary classification; all others to open-ended captioning. The routing is deterministic and error-free by construction, operating on ground-truth answer labels across all 4,793 samples.

4.2 Text-Only Baselines

Questions are vectorized with TF-IDF (10,000 terms, `ngram_range=(1,3)`, `sublinear_tf=True`). We train Logistic Regression (L2, `max_iter=1000`) and SVM (linear kernel) on these features. TF-IDF + SVM achieves 68.95% accuracy and 0.75 AUC-ROC, confirming meaningful but insufficient signal from text alone.

4.3 Image-Only Baseline

ResNet-18 [3] pretrained on ImageNet is fine-tuned with `layer4` unfrozen and a new binary classification head, reusing low-level ImageNet features while adapting high-level representations to the medical domain.

4.4 Multimodal Fusion Models

Our fusion model concatenates image and text features before a learned head:

- **Image encoder**: ResNet-18 (`layer4` unfrozen), 512-dim after global average pooling.
- **Text encoder**:
  Frozen CLIP (512-dim) [8] or BioBERT `dmis-lab/biobert-base-cased-v1.2` [5] (768-dim, mean-pooled last hidden state).
- **Fusion head**: LayerNorm on each stream → concatenation → Linear(256)–ReLU–Dropout(0.5)–Linear(1).

The fused representation is:

$$z_{\text{fused}} = \left[\, \text{LN}(z_{\text{img}}) \oplus \text{LN}(z_{\text{txt}}) \,\right] \quad \text{where } z_{\text{img}} \in \mathbb{R}^{512},\ z_{\text{txt}} \in \mathbb{R}^{768}. \tag{1}$$

BioBERT is chosen over CLIP because it is pretrained on PubMed abstracts and PubMed Central texts, making it better suited to clinical terminology. We additionally fine-tune BioBERT layers 10–11 at 5×10^{-6} to adapt text representations while avoiding catastrophic forgetting.

Models are trained with `BCEWithLogitsLoss` and AdamW with separate LR groups: 10^{-4} (fusion head), 10^{-5} (ResNet `layer4`), 5×10^{-6} (BioBERT layers 10–11). Batch size 16, weight decay 10^{-4}, up to 30 epochs, early stopping (patience 7), `ReduceLROnPlateau` (factor 0.5, patience 3).

4.5 Open-Ended Captioning: GPT-5 Mini

For open-ended questions we query GPT-5 Mini (`gpt-5-mini-2025-08-07`) via the OpenAI API, sending each image as base64 alongside the question prompt.

5 Experiments

5.1 Evaluation Metrics

Binary VQA: accuracy, macro F1, and AUC-ROC on the 959-sample validation split. Open-ended captioning: BLEU-4, ROUGE-L, and BERTScore [10].

5.2 Binary VQA Results

Results show a clear progression from text-only baselines to multimodal fusion. Moving from ResNet-18 alone (72.63%) to CLIP fusion (+3.16 pp) confirms that visual features alone are insufficient without textual grounding. Replacing CLIP with BioBERT adds a further 2.1 pp, reflecting its

Table 1: Binary VQA model comparison on the validation set.

Model	Accuracy	F1	AUC-ROC
TF-IDF + Logistic Regression	0.6684	0.6441	0.7073
TF-IDF + SVM	0.6895	0.6704	0.7517
ResNet-18 (layer4 unfrozen)	0.7263	0.7500	0.7740
Fusion: ResNet-18 + CLIP	0.7579	0.7723	0.8194
Fusion: ResNet-18 + BioBERT	0.7789	0.7835	0.8204
Fusion: ResNet-18 + BioBERT (fine-tuned)	**0.7947**	**0.8000**	**0.8281**

stronger clinical vocabulary alignment. Fine-tuning the top two BioBERT layers provides the final boost to 79.47% accuracy and macro-F1 of 0.80.

5.3 Open-Ended Captioning Results

Table 2: Open-ended captioning results (GPT-5 Mini) on the validation set.

Model	BLEU-4	ROUGE-L	BERTScore
GPT-5 Mini (zero-shot)	–	–	–
GPT-5 Mini (few-shot)	–	–	–

5.4 Qualitative Analysis

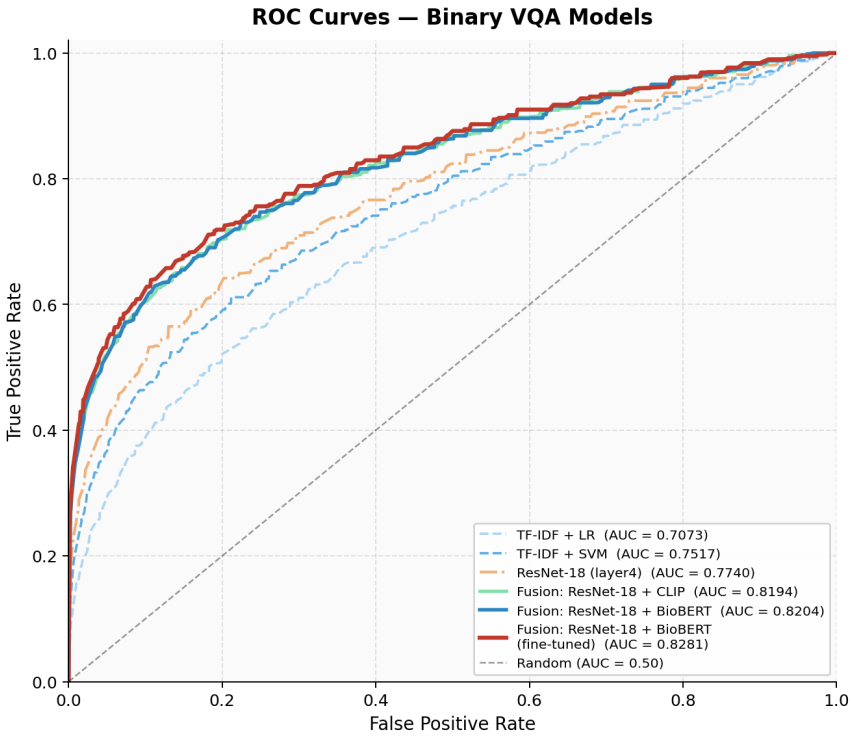

Figure 2: ROC curves for binary VQA models. The fine-tuned BioBERT fusion (AUC = 0.8281) consistently outperforms the vision-only ResNet-18 baseline (AUC = 0.7740).

5.5 Compute Resources

Binary fusion experiments used NVIDIA T4 and A100 GPUs via Google Colab Pro, requiring ≈45 minutes to converge over 30 epochs (batch size 16, max sequence length 128). Open-ended evaluation used the OpenAI API.

6 Conclusion

We presented MedVQA-Router, a two-stage pipeline for Medical VQA that routes questions to a binary classification branch or an open-ended captioning branch. Combining ResNet-18 with fine-tuned BioBERT achieves 79.5% accuracy and 0.83 AUC-ROC, outperforming all single-modality baselines by over 7 pp. The consistent advantage of BioBERT over CLIP confirms the value of domain-adapted language models for medical text understanding. Future work includes cross-attention fusion for finer-grained image-text interaction, end-to-end multi-task routing, and evaluation on larger benchmarks such as VQA-RAD and SLAKE [7].

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